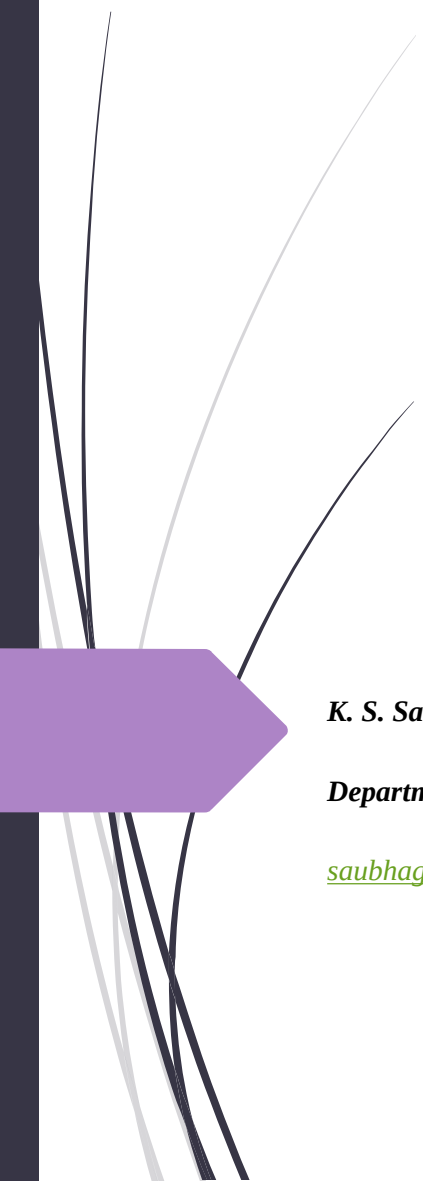




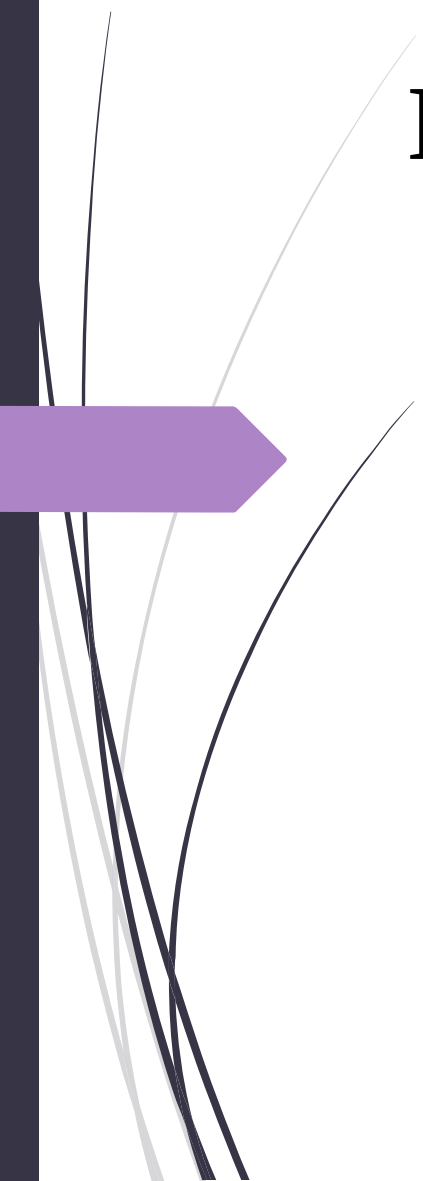
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Operational Research

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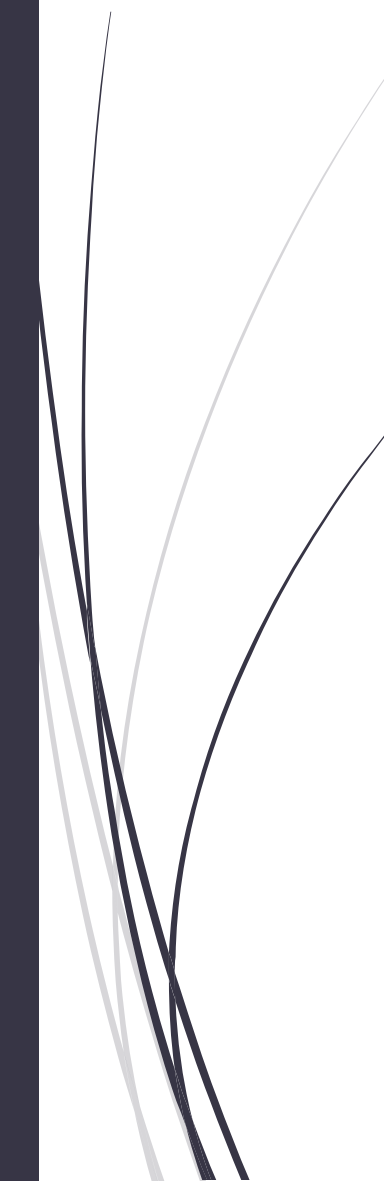


Binary Integer Programming (Zero-one Programming)

Chapter 3



Chapter Summary



- **Introduction to Binary Integer Programming**
- **Logical Constraints - multiple-choice constraints**
- **Logical Constraints – Either-Or Constraint**
- **K out of N constraints must hold**
- **At least K out of N constraints must hold**
- **Logical Constraints – If-then constraint**
- **Binary representation of general integer variables**

Introduction

- In previous lessons we discussed about Mixed Integer Programming (MIP) problems and Pure Integer Programming problems (PIP).
- There is another area of application of integer programming; problems involving a number of inter-related “yes or no decisions”. In such decisions, there is only two possible choices, “yes” or “no”.
- For example, suppose that we are to determine whether or not to engage in the following activities:
 - Should we build a new plant in a particular location?
 - Should we undertake an advertising campaign?
 - Should we develop a new product?
- In each of above case, we must make a **yes–no** or so-called **go–no–go** decision.

Binary (0-1) Variables

- ▶ With just two choices, we can represent such decisions by decision variables that are restricted to just two values, 0 or 1.
- ▶ Variables that are restricted to 0 or 1 in this way are termed binary, bivalent, logical, or 0-1 variables.

$$X_i = \begin{cases} 1, & \text{if decision } i \text{ is yes} \\ 0, & \text{if decision } i \text{ is no} \end{cases}$$

- ▶ Binary variables are of great importance because they occur regularly in many model formulations, particularly in problems addressing long-range and high-cost strategic decisions.
- ▶ Integer Programming problem that only contain binary variables are called Binary Integer Programming (BIP) problems.
- ▶ Note that, there may be problems that contain both binary and other variables.

Example – Capital Budgeting

- ▶ Decisions about whether or not to undertake a project is usually made under limited budget considerations and preset priorities.
- ▶ E.g.: Consider a situation where five projects are being evaluated over a 3-year planning horizon. The following table gives the expected returns for each project and the associated yearly expenditures:

Project	Expenditure (\$ million)/year			Returns (\$ million)
	1	2	3	
1	5	1	8	20
2	4	7	10	40
3	3	9	2	20
4	7	4	1	15
5	8	6	10	30
Available funds (\$ million)	25	25	25	

- ▶ Which projects should be selected over the 3-year horizon?

Example – Capital Budgeting

- ▶ The problem reduces to a “yes–no” decision for each project. Define the binary variable X_i as

$$X_i = \begin{cases} 1, & \text{if project } i \text{ is selected} \\ 0, & \text{if project } i \text{ is not selected} \end{cases}$$

- ▶ The ILP model is

$$\text{Maximize } Z = 20X_1 + 40X_2 + 20X_3 + 15X_4 + 30X_5$$

$$\text{Subject to; } 5X_1 + 4X_2 + 3X_3 + 7X_4 + 8X_5 \leq 25$$

$$X_1 + 7X_2 + 9X_3 + 4X_4 + 6X_5 \leq 25$$

$$8X_1 + 10X_2 + 2X_3 + X_4 + 10X_5 \leq 25$$

$$X_1, X_2, X_3, X_4, X_5 = (0,1)$$

- ▶ The optimum integer solution is $X_1 = X_2 = X_3 = X_4 = 1, X_5 = 0$, with $Z = 95$ (\$ million).
- ▶ Note that the solution excludes project 5 from the product mix.

Logical Constraints - multiple-choice constraints

- ▶ The capital-budgeting model can be made much richer by including logical considerations.
- ▶ Suppose, for example, that project 1 is contingent upon project 2. This contingency is modeled simply by the constraint $X_2 \geq X_1$.
- ▶ Constraints like this are commonly called multiple-choice constraints.

Examples of multiple-choice constraints:

- ▶ Exactly one project should be selected from the set of projects P1, P2 and P3: $X_1 + X_2 + X_3 = 1$
- ▶ At least one project should be selected from the set of projects P1,P2 and P3: $X_1 + X_2 + X_3 \geq 1$
- ▶ At most one project should be selected from the set of projects P1,P2 and P3: $X_1 + X_2 + X_3 \leq 1$

Logical Constraints - multiple-choice constraints

Examples of multiple-choice constraints continued:

- ▶ At most one of the two projects P2 and P4 should be selected:
 $X_2 + X_4 \leq 1$
- ▶ If P3 is selected, then P4 must be selected: $X_3 \leq X_4$
- ▶ If P1 and P2 are both selected, then P5 should be selected:
 $X_1 + X_2 - X_5 \leq 1$

Logical Constraints – Either-Or Constraint

- ▶ Consider the case where only one of the two constraint should hold, either the first one or the second one.
- ▶ For example, suppose that there are two production process that could be used to produce a set of products in a manufacturing company. The technological coefficients (ex- amount of raw material) required for products may vary and the amount available also may vary for two production processes. Manufacturer wants to select only one production process not both.
- ▶ As for an illustration, consider two constraints;
$$3X_1 + 2X_2 \leq 18$$
$$X_1 + 4X_2 \leq 16$$
- ▶ How can we include the above restriction where we want only one constraint depending on the production process used?

Logical Constraints – Either-Or Constraint

- Solution is to add a large positive number to the Right-Hand Side (RHS) of the inequality, which has the effect of eliminating that restriction.
- When you add a large positive number to RHS, it means you have that resource in abundance so that there would not be any restriction of that resource. So, the constraint is eliminated.

- Let's say this large positive number is M and it is added to the constraints;

Either $3X_1 + 2X_2 \leq 18 + M$

Or $X_1 + 4X_2 \leq 16 + M$ should occur

- To make that only one constraint happens at a time, we can modify the constraint using a binary variable Y as below;

$$\begin{aligned} 3X_1 + 2X_2 &\leq 18 + MY \\ X_1 + 4X_2 &\leq 16 + M(1 - Y) \end{aligned}$$

- If $Y = 0$, constraint one will occur and otherwise constraint two occurs.

K out of N constraints must hold

- Let's assume we have N constraints;

$$f(X_1, X_2, \dots, X_n) \leq d_j \text{ for } j = 1, 2, \dots, N$$

- How do we take only K constraints out of N constraints?
- Let's add a large positive number M to all the N constraints and use a binary variables Y_j s for all constraints as below.

$$f(X_1, X_2, \dots, X_n) \leq d_1 + MY_1$$

$$\vdots \qquad \qquad \qquad \vdots$$

$$f(X_1, X_2, \dots, X_n) \leq d_N + MY_N$$

- To take only K constraints we add another constraint on summation of Y_j s such that;

$$\sum_{j=1}^N Y_j = N - K$$

At least K out of N constraints must hold

- In the above situation, we want to take exactly K number of constraints, however if we want to take at least K number of constraints we can amend the constraint with a multiple-choice constraint as follows;

$$\sum_{j=1}^N Y_j \leq N - K$$

- For example, consider situation where we have two constraints. Then at least one, but not necessarily both, of these constraints must be valid. This restriction can be modeled as below.

$$f(X_1, X_2, \dots, X_n) \leq d_1 + MY_1$$

$$f(X_1, X_2, \dots, X_n) \leq d_2 + MY_2$$

$Y_1 + Y_2 \leq 1$ where Y_1 and Y_2 are binary

Logical Constraints – If-then constraint

- Suppose that we have a condition where one constraint should precede another, then we have if-then condition. Mathematically, the if-then condition is written as below.

$$\text{If } f(X_1, X_2, \dots, X_n) \leq d_1, \text{ then } f(X_1, X_2, \dots, X_n) \leq d_2$$

- Given $\varepsilon(> 0)$ infinitesimally small, and M sufficiently large, this condition is equivalent to the following two simultaneous constraints:

$$d_1 - f(X_1, X_2, \dots, X_n) \leq M(1 - Y) - \varepsilon$$

$$f(X_1, X_2, \dots, X_n) - d_2 \leq MY$$

where Y is binary.

- If $f(X_1, X_2, \dots, X_n) \leq d_1$ then, $d_1 - f(X_1, X_2, \dots, X_n) \geq 0$ which requires $Y = 0$, and the second constraint becomes $f(X_1, X_2, \dots, X_n) \leq d_2$ as desired. Else, Y may assume the value 0 or 1, in which case the second constraint may or may not be satisfied, depending on other conditions in the model.

Binary representation of general integer variables

- ▶ If there is an IP problem where most of the variables are Binary, then it is convenient to convert it to a pure BIP problem.
- ▶ Let X be an integer variable bounded by the value u , $0 \leq X \leq u$.
- ▶ Find N such that $2^N \leq u < 2^{N+1}$.
- ▶ The binary representation of X is $X = \sum_{i=0}^N 2^i Y_i$ where Y_i s are binary variables.

Example: Binary representation of general integer variables

- Consider the following constraints:

$$X_1 \leq 5 \text{ and } 2X_1 + 3X_2 \leq 30$$

- These constraints imply that $u = 5$ for X_1 and $u = 10$ for X_2 .

$$2^2 < 5 < 2^3 \text{ and } 2^3 < 10 < 2^4$$

$$N = 2 \text{ for } X_1 \text{ and } N = 3 \text{ for } X_2$$

$$X_1 = Y_0 + 2Y_1 + 4Y_2 \text{ and } X_2 = Y_3 + 2Y_4 + 4Y_5 + 8Y_6$$

- So, the constraints with binary variables are;

$$Y_0 + 2Y_1 + 4Y_2 \leq 5$$

$$2(Y_0 + 2Y_1 + 4Y_2) + 3(Y_3 + 2Y_4 + 4Y_5 + 8Y_6) \leq 30$$

Exercise 01: Fixed-Charge Problem

- Service charges of three telephone companies for long distance calls are given as follows.

Company	Flat Monthly Charge (\$)	Cost/minute (\$)
1	16	0.25
2	25	0.21
3	18	0.22

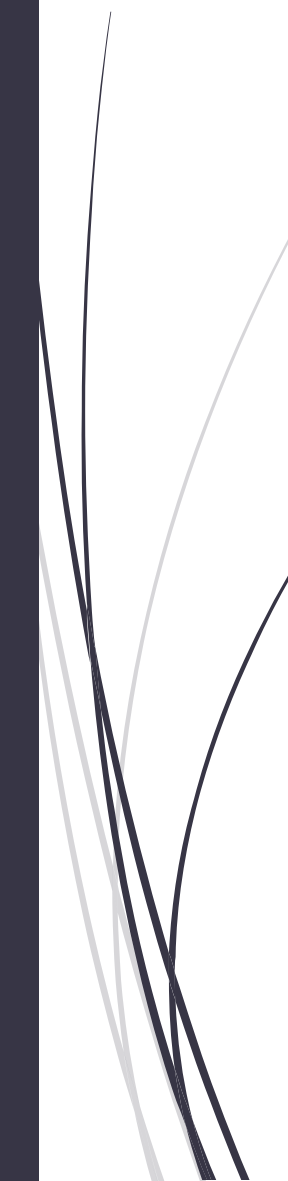
- A person usually makes on average 200 minutes of long-distance calls per month.
- Assuming that, no flat monthly fee payment unless make calls, person can apportion calls among all three companies as he/she please, how should a person use the three companies to minimize the monthly telephone bill?



Exercise 01: Fixed-Charge Problem Solution

- ▶ The fixed-charge problem deals with situations in which the economic activity incurs two types of costs: a fixed cost needed to initiate the activity and a variable cost proportional to the level of the activity.

Solution:



Exercise 02

- ▶ A Company is considering two potential drilling sites for reaching four targets (possible oil wells). The following table provides the preparation costs at each of the two sites and the cost of drilling from site i to target j ($i = 1, 2; j = 1, 2, 3, 4$).

Site	Well				Fixed Preparation cost (million \$)
	1	2	3	4	
1	2	1	8	5	5
2	4	6	3	1	6

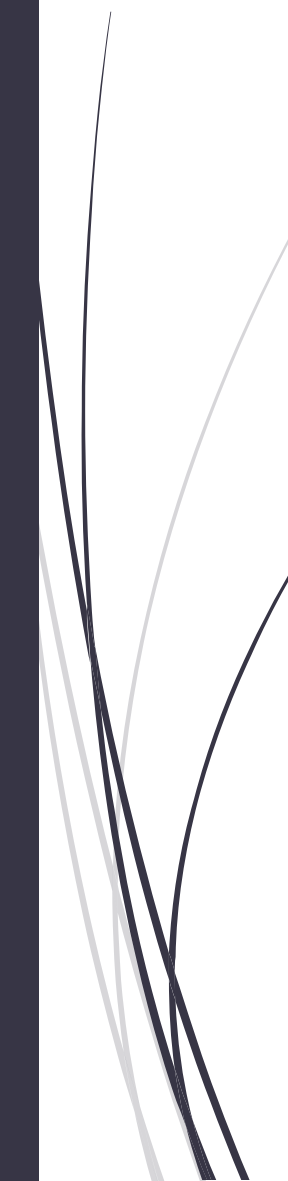
- ▶ Clearly define the decision variables for this BIP problem and formulate the objective function and corresponding constraints.
- ▶ Any oil well should receive oil from any one of the two sites, either from the first one or the second one, not from both.
- ▶ For site 1, all the 4 wells may be connected. But for site 2, the officers decide to connect maximum of 3 wells.



Exercise 02: Solution

- Note that for this situation, decisions must be made about tradeoffs between drilling costs and costs for operating drilling centers.

Solution:



Exercise 03

- ▶ Young couple Eve and Steve wants to divide their main household activities (cleaning, cooking, dishwashing and laundering) between them so that each has two tasks. However, the total time they spend on household duties is kept to a minimum. Their efficiencies on these tasks differ, where the time each would need to perform the tasks is given by the below table.

	Time needed per week (in hours)			
	Cleaning	Cooking	Dishwashing	Laundry
Eve	4.5	7.8	3.6	2.9
Steve	4.9	7.2	4.3	3.1

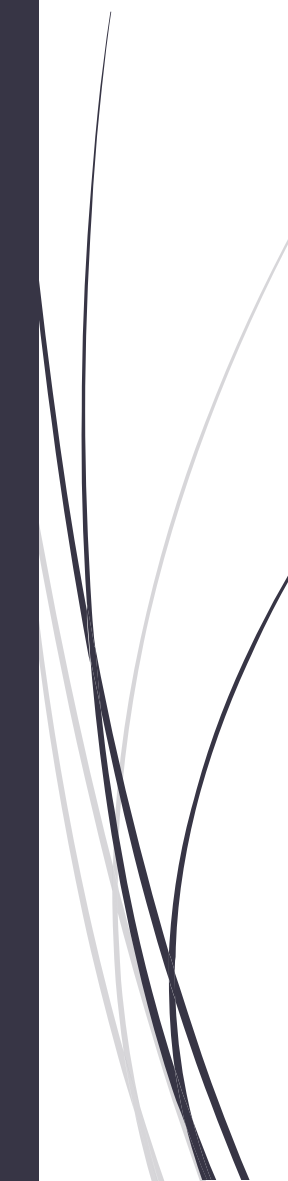
- ▶ Define the required variable and formulate the BIP problem.

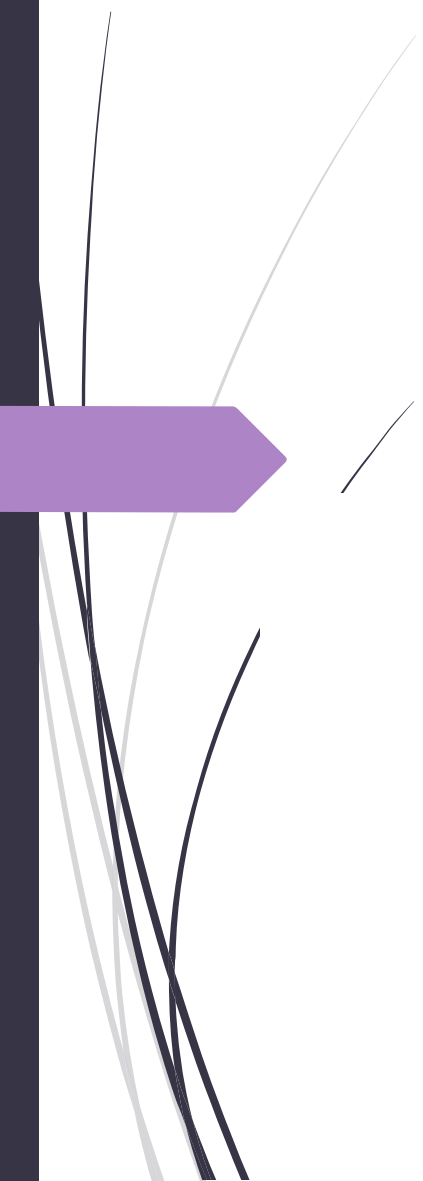


Exercise 03: Solution

- Decisions should be made to reduce the time spent in household activities with other constraints (all activities should be done and two activities per person)

Solution:





Thank You!

Next Chapter - Transportation Model